Lawrence R. Liu

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Education

University of California Los Angeles — Los Angeles, CA
MS/PhD, Electrical Engineering, GPA: 4.0/4.0 — *Sep 2024 – June 2029*

University of California Los Angeles — Los Angeles, CA
BS, Electrical Engineering, GPA: 3.8/4.0 — *Sep. 2021 – June 2024*

Research Interests

Sparse structure and compression of large language models (pruning, quantization, matrix factorization), reinforcement learning for LLM optimization, applied machine learning for clinical neuroscience.

Research Experience

Research Assistant — LLM Sparsity, Compression & Reasoning — April 2023 – Present
Advised By Professor Lin F. Yang — *UCLA*

- Developed **ARMOR**, a semi-structured (2:4) LLM pruning algorithm via adaptive matrix factorization of sparse cores with block-diagonal error correctors; proved convergence guarantees via block coordinate descent
- **ARMOR** outperforms SOTA 2:4 pruning by up to **+10 pts on MMLU** and **+17 pts on GSM8K** (Qwen-3-8B); reduces the C4 perplexity gap to dense by ∼**50%** on Llama-3-70B
- Developed **NoWag**, a unified normalization and proxy-loss framework for data/compute-efficient LLM pruning and quantization; a single framework matches or surpasses compression-paradigm-specific SOTA methods
- **NoWag-VQ** surpasses all SOTA 2-bit vector quantization baselines across Llama-2 (7B/13B/70B) and Llama-3 8B using **48× less calibration data**
- Conducted experiments and theoretical analysis on **LACONIC**, a primal-dual Lagrangian-based length-constrained RL framework for LLM reasoning optimization; empirically demonstrated >**50%** reduction in reasoning length while maintaining or improving accuracy on MATH and OlympiadBench

Research Assistant — AI for Neuroscience — February 2022 – October 2024
Advised By Professor Vwani Roychowdhury — *UCLA*

- Leveraged self-supervised deep learning (**VAE**) to automatically discover and cluster novel morphologies in epileptic brain signals, enabling systematic scientific characterization of pathological activity
- ML-driven latent-space analysis revealed a new class of high-frequency oscillations more predictive of surgical outcomes than established biomarkers; validated across **686K events from 185 patients**, outperforming the clinical SOZ-based standard (F1=0.83 vs. 0.74)
- Built **PyHFO**, an open-source AI platform for clinical neuroscience combining optimized signal detection with lightweight CNN classifiers; achieved **50× speedup** over standard methods and deployed to UCLA Mattel Children's Hospital

Publications

Journal & Conference Papers

- **Lawrence Liu**, Alexander Liu, Mengdi Wang, Tuo Zhao, Lin F. Yang, "ARMOR: High-Performance Semi-Structured Pruning via Adaptive Matrix Factorization", *International Conference on Learning Representations (ICLR)*, 2026.
- **Lawrence Liu**, Inesh Chakrabarti, Yixiao Li, Mengdi Wang, Tuo Zhao, Lin F. Yang. "NoWag: A Unified Framework for Shape Preserving Compression of Large Language Models", *Conference on Language Modeling (COLM)*, 2025.
- Yipeng Zhang, Atsuro Daida, **Lawrence Liu**, Naoto Kuroda, Yuanyi Ding, Shingo Oana, Sotaro Kanai, Tonmoy Monsoor, Chenda Duan, Shaun A. Hussain, Joe X. Qiao, Noriko Salamon, Aria Fallah, Myung Shin Sim, Raman Sankar, Richard J. Staba, Jerome Engel Jr., Eishi Asano, Vwani Roychowdhury, Hiroki Nariai. "Self-Supervised Data-Driven Approach Defines Pathological High-Frequency Oscillations in Epilepsy", *Epilepsia*, 2025.
- Yipeng Zhang, **Lawrence Liu**, Yuanyi Ding, Xin Chen, Tonmoy Monsoor, Atsuro Daida, Shingo Oana, Shaun Hussain, Raman Sankar, Aria Fallah, Cesar Santana-Gomez, Jerome Engel, Richard J. Staba, William Speier, Jianguo Zhang, Hiroki Nariai, Vwani Roychowdhury. "PyHFO: Lightweight Deep Learning-powered End-to-End High-Frequency Oscillations Analysis Application", *Journal of Neural Engineering*, 21, 2024.

Preprints & Under Review

- Chang Liu, Yiran Zhao, **Lawrence Liu**, Yaoqi Ye, Csaba Szepesvari, Lin F. Yang, "LACONIC: Length-Aware Constrained Reinforcement Learning for LLM", 2026.
- Ramnath Kumar, Kyle Ritscher, Junmin Zhu, **Lawrence Liu**, Cho-Jui Hsieh, "Flex-Act: Why Learn when you can Pick?", 2026.

- Tonmoy Monsoor, Sotaro Kanai, Atsuro Daida, Naoto Kuroda, Prateik Sinha, Shingo Oana, Yipeng Zhang, **Lawrence Liu**, Gaurav Singh, Chenda Duan, Myung Shin Sim, Aria Fallah, William Speier, Eishi Asano, Vwani Roychowdhury, Hiroki Nariai. "Mini-Seizures: Novel Interictal iEEG Biomarker Capturing Synchronization Network Dynamics at the Epileptogenic Zone", medRxiv, 2025.

Workshop Papers

- Preliminary results of the NoWag framework presented at **ICLR 2025 Sparsity in LLMs Workshop**.

Teaching Experience

Teaching Assistant — April 2024 – June 2024
UCLA Electrical and Computer Engineering Department — *Los Angeles, CA*

- Served as TA for graduate-level Deep Learning 2 (ECE 239AS) as an undergraduate, leading weekly office hours and RL-focused discussion sections for 200+ graduate students
- Designed two reinforcement learning programming assignments (DDPG for continuous control, DQN for discrete actions) adopted for course curriculum, enabling hands-on implementation experience with modern RL algorithms

Awards, Honors & Services

Dean's List — *UCLA Henry Samueli School of Engineering* — Fall 2021, Fall 2022, Winter 2023, Spring 2023, Spring 2024
Eta Kappa Nu (HKN) Honor Society — *UCLA* — Fall 2023
Services — Reviewer: ICLR 2026 Mentor: UCLA ACM AI AISF Program

Technical Skills

Languages: Python, C/C++, R, Matlab, Octave, Julia, Mathematica
Frameworks & Tools: Git, Bash Scripting, Pytorch, Sklearn, Tensorflow, Pandas, NumPy, Matplotlib, Simulink, Gymnasium, Optuna, Cvxpy, Mosek, Cuda, Transformers
Machine Learning: Deep Learning, Reinforcement Learning, Graph Neural Networks, Generative AI, LLM Compression, LLM Evaluation, Inference Optimization, Model Parallelism
Signal Processing: Time Series Analysis, Statistical Analysis, Data Mining, Data Visualization
Optimization: Numerical Analysis, Convex Optimization
Systems: Control Theory, High-Performance Computing